

WHITEPAPER

Engineering Long-Life Battery-Powered Smart Water Meters

Part 6 — From Laboratory to Field: Achieving Real-World Battery Life

A Systems Engineering Approach to Battery Life in Advanced Metering Infrastructure

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Part	6 of 7 — From Laboratory to Field: Achieving Real-World Battery Life Whitepaper Series: ☐ Part 1 Understanding Battery Life and the Battery Life Budget ☐ Part 2 Battery Chemistry and Energy Storage ☐ Part 3 Ultra-Low-Power Hardware Architecture ☐ Part 4 Firmware as the Energy Manager ☐ Part 5 Battery Life Qualification and Validation ☒ Part 6 From Laboratory to Field: Achieving Real-World Battery Life ☐ Part 7 Practical Recommendations and the Future of Long-Life Smart Water Metering
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“The laboratory estimates battery life. The field reveals it.”

6.1 The Gap Between Design and Deployment

One of the most common questions raised during AMI projects is:

"If the meter was designed for fifteen years, why is the battery lasting much less?"

The answer is rarely simple.

Battery life predictions are developed using engineering assumptions that define how the product is expected to operate.

These assumptions typically include:

- Reporting frequency
- Communication success rate
- Expected signal quality
- Ambient temperature
- Firmware behaviour
- Battery characteristics
- Normal operating conditions

During actual deployment, however, many of these assumptions change.

As operating conditions diverge from the original Battery Life Budget, actual battery consumption also changes.

Understanding this gap is essential for both manufacturers and utilities.

Engineering Predictions Depend on Assumptions

Battery Life Budgets are built using measurable engineering inputs.

Typical assumptions include:

- One successful transmission per day
- Minimal communication retries
- Stable network coverage
- Average operating temperature
- Defined wake-up schedule
- Expected MCU active time
- Known battery capacity

When these assumptions remain valid, battery life predictions are generally reliable.

When deployment conditions differ significantly, battery performance will naturally change.

This does not necessarily indicate a defective battery.

It often reflects a difference between the assumed operating environment and actual field conditions.

Real Deployments Are Dynamic

Unlike laboratory testing, field deployments evolve continuously.

Conditions may change due to:

- Cellular network optimisation
- New base station configurations
- Urban development
- Vegetation growth
- Infrastructure changes
- Customer behaviour
- Firmware upgrades
- Utility operational policies

Each of these factors may influence energy consumption over the operational life of the meter.

Battery performance should therefore be viewed as dynamic rather than fixed.

Battery Life Is a System Outcome

Utilities sometimes evaluate battery life as though it were solely determined by the battery.

In reality, field performance reflects the behaviour of the complete AMI system.

Examples include:

- RF coverage
- Communication technology
- Firmware strategy
- Data collection policy
- Installation quality
- Environmental exposure
- Utility operational procedures

Changing any one of these elements may influence overall battery consumption.

Consequently, battery life should be managed as a system-level performance indicator rather than as an isolated battery specification.

Utilities Influence Battery Life Too

Battery performance does not depend exclusively on product design.

Utility operational decisions also affect long-term energy consumption.

Examples include:

- Increasing reporting frequency
- Requesting additional diagnostics

- Expanding alarm functionality
- Performing more frequent remote reads
- Scheduling firmware updates
- Changing communication policies

Each operational decision consumes part of the available Battery Life Budget.

Understanding these trade-offs allows utilities to balance operational value against long-term service life.

Design Margin Exists for a Reason

Experienced engineering teams intentionally include battery margin within the Battery Life Budget.

This reserve helps accommodate:

- Component ageing
- Manufacturing variation
- Environmental uncertainty
- Communication variability
- Future firmware enhancements
- Operational flexibility

However, this margin is finite.

Repeated changes that increase energy consumption without considering their cumulative impact may gradually reduce the expected operational lifetime.

Managing this margin carefully is therefore an important part of long-term AMI operation.

Engineering Insight

Battery life predictions are not guaranteed. They are engineering estimates based on defined assumptions. The closer field operation remains to those assumptions, the closer actual battery life will match the original design target.

Practitioner Observation

Utilities occasionally interpret shorter-than-expected battery life as evidence of battery quality issues. In practice, field investigations often reveal that increased reporting frequency, poorer-than-anticipated RF conditions, additional communication retries, or evolving operational requirements have increased energy consumption beyond the assumptions used during product qualification. Reviewing these operational factors is frequently more informative than focusing on the battery alone.

Key Takeaways

- Battery life predictions depend on clearly defined engineering assumptions.
- Field deployments evolve over time and rarely remain identical to laboratory conditions.
- Battery performance reflects the behaviour of the complete AMI system.
- Utility operational decisions directly influence long-term energy consumption.
- Battery margin should be managed as a valuable engineering resource rather than unused capacity.

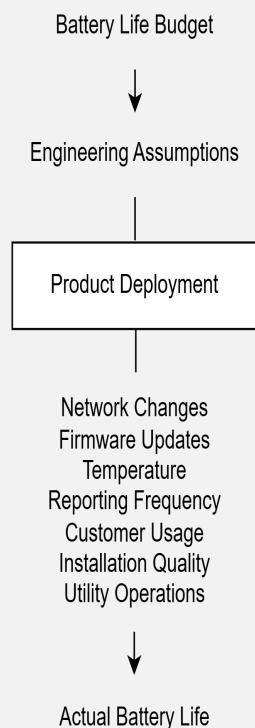


Figure 6.1 — Why Battery Life Changes After Deployment

Battery life predictions are based on engineering assumptions. As deployment conditions evolve, actual battery performance reflects the combined influence of network, environment, firmware, installation quality, and utility operational decisions.

6.2 How Network Quality Changes Battery Life

Communication is one of the most energy-intensive activities performed by a smart water meter.

During normal operation, the device spends the vast majority of its life in an ultra-low-power sleep state. Battery energy is consumed primarily when the meter wakes to measure consumption, process data, and communicate with the AMI network.

As a result, communication quality has a direct influence on long-term battery performance.

Reliable networks minimise transmission time and reduce unnecessary retries, allowing the meter to return to sleep quickly. Poor network conditions have the opposite effect, increasing active operating time and consuming additional Battery Life Budget.

Retries Consume More Energy Than Successful Transmissions

One successful transmission is generally less expensive than multiple failed attempts.

If the communication channel is unreliable, the device may repeatedly:

- Wake the processor
- Power the radio
- Search for the network
- Attempt authentication
- Retransmit data
- Wait for acknowledgement

Each retry consumes additional battery energy.

While occasional retries are expected, persistent communication failures can significantly reduce operational lifetime.

Battery qualification should therefore consider realistic retry behaviour rather than assuming every transmission succeeds on the first attempt.

Network Conditions Change Over Time

Communication quality should not be regarded as fixed.

Throughout the operational life of a smart water meter, network performance may change due to:

- Infrastructure upgrades
- New base stations
- Network optimisation
- Urban development
- Vegetation growth
- Building construction
- Operator configuration changes

A deployment that performs well during initial installation may therefore experience different communication characteristics several years later.

Battery life management should recognise that communication conditions evolve throughout the product lifecycle.

Communication Policy Matters

Battery consumption is influenced not only by network quality but also by communication policy.

Examples include:

- Reporting frequency
- Alarm transmission strategy
- Data packet size
- Time synchronisation intervals
- Remote diagnostics
- Firmware update scheduling

More frequent communication generally provides greater operational visibility but also consumes more battery energy.

Utilities should therefore balance operational requirements against long-term battery objectives.

Design for Communication Efficiency

Engineering teams can improve battery performance by designing communication strategies that minimise unnecessary radio activity.

Examples include:

- Transmitting only meaningful data
- Aggregating multiple measurements into a single message
- Avoiding unnecessary network attachment
- Optimising retry algorithms
- Scheduling communication during appropriate operating windows
- Returning to sleep immediately after successful transmission

Reducing communication overhead often produces greater battery life improvements than further optimisation of already efficient sleep current.

Monitor Communication Performance During Operation

Battery management should continue after deployment.

Useful operational indicators include:

- Daily data availability
- Communication success rate
- Average retry count
- Upload latency
- Missed reporting events
- Network attachment failures

Monitoring these indicators helps utilities identify communication problems before they begin affecting battery performance across large meter populations.

Engineering Insight

The battery does not know why the radio remains active. It only supplies energy. Every additional second spent communicating consumes part of the Battery Life Budget.

Practitioner Observation

Field investigations frequently show that unexpectedly high battery consumption is associated with communication behaviour rather than battery defects. Devices installed in areas with weaker coverage or higher interference may require more retries, longer network attachment times, or repeated reconnection attempts. Monitoring communication quality alongside battery health provides a more complete understanding of long-term field performance than evaluating battery voltage alone.

Key Takeaways

- Communication is one of the largest contributors to battery energy consumption.
- Reliable networks reduce transmission time and minimise retries.
- Communication quality evolves throughout the deployment lifecycle.
- Utility communication policies directly influence battery life.
- Monitoring network performance supports proactive battery management.

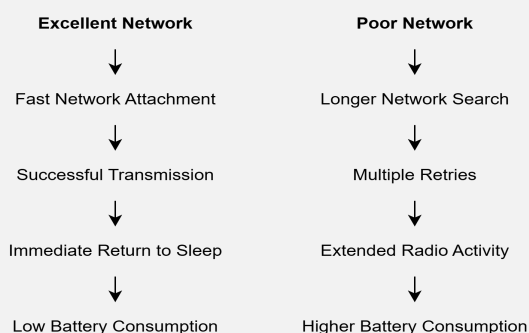


Figure 6.2 — Relationship Between Network Quality and Battery Consumption

Communication quality directly influences battery consumption. Reliable network conditions minimise radio active time, while poor communication quality increases retries and prolongs energy-intensive transmission events.

Battery Engineering Perspective

This chapter introduces an important operational concept:

Utilities should monitor communication efficiency as a leading indicator of battery health.

Rather than waiting for battery voltage to decline, utilities can proactively monitor metrics such as:

- Communication success rate
- Average retries per upload
- Daily data availability
- Network attachment time
- Upload latency
- RF signal quality trends

These indicators often reveal emerging issues months or even years before measurable battery degradation becomes apparent.

6.3 Installation Practices That Increase Power Consumption

Smart water meters are designed and qualified under defined engineering assumptions.

However, actual battery life depends not only on product design but also on how the device is installed.

Two identical meters manufactured on the same production line may experience significantly different battery performance simply because they are installed in different environments or using different installation practices.

Installation quality therefore forms an important part of long-term battery management.

Installation Is Part of the Battery Life Budget

During product development, engineers estimate battery life assuming representative installation conditions.

Once deployed, however, installation quality influences several aspects of system operation, including:

- RF communication performance
- Environmental exposure
- Antenna efficiency
- Temperature
- Moisture ingress
- Accessibility for maintenance

Poor installation practices often increase energy consumption indirectly by making communication more difficult rather than increasing electrical current directly.

Underground Meter Chambers

One of the most challenging deployment environments for smart water meters is the underground meter chamber.

These installations frequently present:

- Reinforced concrete walls
- Metal covers
- High moisture
- Flooding
- Restricted RF propagation

These conditions may reduce communication efficiency, increasing:

- Network attachment time
- Retry frequency
- Radio active duration

Consequently, battery consumption may increase even though the meter itself is operating correctly.

Where underground deployment is unavoidable, communication performance should be verified during installation.

Antenna Placement Matters

Wireless performance depends heavily on antenna effectiveness.

Poor antenna installation may result from:

- Incorrect orientation
- Obstruction by metal structures
- Damage during installation
- Excessively tight cable routing
- Physical deformation

Even modest reductions in antenna efficiency may require the communication module to remain active longer before successful data transmission.

Maintaining antenna integrity is therefore essential for preserving battery performance.

Environmental Exposure

Installation location also determines the environmental conditions experienced by the meter.

Examples include:

- Direct sunlight
- Standing water
- High humidity
- Roadside heat
- Poor ventilation
- Chemical exposure

These conditions may accelerate battery ageing or influence communication reliability throughout the product's operational life.

Installation guidelines should therefore consider environmental protection in addition to mechanical security.

Installation Workmanship

The quality of installation directly affects long-term system reliability.

Examples of poor workmanship include:

- Loose electrical connections
- Damaged seals
- Incorrect enclosure assembly
- Inadequate cable protection
- Improper torque on sealing components
- Mechanical stress on connectors

Such issues may not produce immediate failures but can gradually reduce product reliability and battery performance over time.

Standardised installation procedures and technician training help minimise these risks.

Verify Communication After Installation

Functional testing should not end when the meter is physically installed.

Commissioning procedures should confirm that the installed device achieves acceptable communication performance.

Useful commissioning checks include:

- Successful network registration
- Signal quality verification
- Initial data upload
- Alarm transmission test
- Time synchronisation

- Battery voltage recording

These baseline measurements provide valuable reference data for future maintenance and troubleshooting.

Installation Data Should Be Retained

Utilities often focus on billing data after deployment while overlooking installation records.

Retaining installation information such as:

- GPS location
- Installation date
- Signal quality at commissioning
- Chamber type
- Antenna orientation
- Installer identification

can greatly simplify later investigations into battery performance or communication issues.

Good documentation transforms isolated failures into valuable engineering insights across the entire deployment.

Engineering Insight

The battery powers the meter, but the installation environment determines how efficiently that energy is used. Good installation practices preserve the Battery Life Budget; poor installation silently consumes it.

Practitioner Observation

Field investigations frequently show that meters with unexpectedly high battery consumption are not defective. Instead, they are often installed in locations where RF propagation is poor, underground chambers remain waterlogged, or antenna performance has been compromised during installation. Improving installation quality frequently delivers greater battery life improvements than replacing the battery itself.

Key Takeaways

- Installation quality directly influences long-term battery performance.
- Underground chambers and obstructed RF environments increase communication energy.
- Proper antenna placement supports efficient wireless communication.
- Environmental exposure should be considered during installation planning.
- Commissioning measurements provide valuable baseline data for future diagnostics.

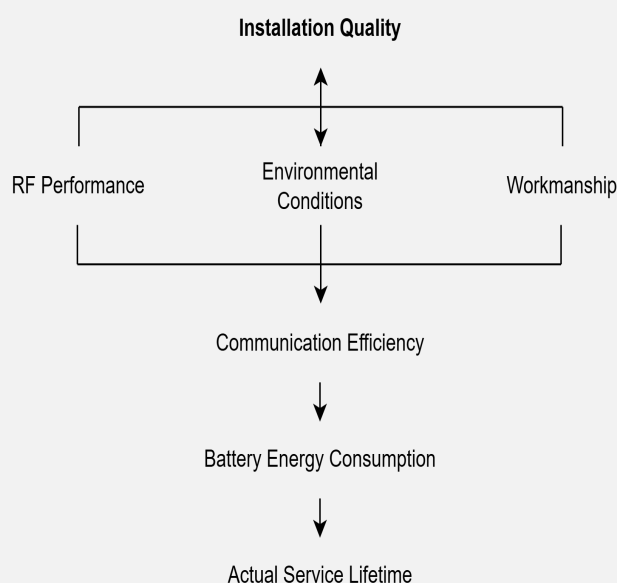


Figure 6.3 — Installation Factors Influencing Battery Life

Installation quality influences battery life through its effects on communication efficiency, environmental exposure, and long-term system reliability. Consistent installation practices help ensure that deployed meters operate within the assumptions of the original Battery Life Budget.

6.4 Utility Operational Policies and Their Battery Impact

Once a smart water meter has been deployed, battery life is no longer determined solely by product design.

Daily operational decisions made by the utility continue to influence how quickly battery energy is consumed.

Reporting schedules, alarm strategies, firmware management, remote diagnostics, and data collection policies all affect the frequency and duration of communication events.

Although each individual decision may have only a small impact, their cumulative effect over many years can significantly alter the actual operational lifetime of the meter.

Effective battery management therefore requires both sound engineering design and disciplined operational governance.

Every Communication Event Has an Energy Cost

Every time a smart meter communicates, it consumes battery energy.

Typical communication events include:

- Scheduled meter readings
- Alarm notifications
- On-demand remote reads
- Configuration updates
- Time synchronisation
- Firmware downloads
- Network re-registration
- Diagnostic requests

While these functions provide valuable operational information, they also increase total battery consumption.

Utilities should therefore understand the energy implications of each operational activity.

Reporting Frequency Is a Strategic Decision

One of the most influential operational settings is the reporting interval.

For example:

- One transmission per day
- Four transmissions per day
- Hourly reporting
- Near real-time reporting

More frequent reporting provides greater operational visibility but also increases communication activity.

The appropriate reporting interval should therefore be determined by operational objectives rather than by technology capability alone.

For many applications, increasing reporting frequency beyond operational requirements provides limited additional value while consuming more of the Battery Life Budget.

Alarm Configuration Should Be Purpose-Driven

Modern smart water meters support numerous alarm functions.

Examples include:

- Leak detection
- Reverse flow
- Continuous flow
- Tamper detection
- Low battery warning
- Communication failure
- Burst detection

Although these alarms improve operational awareness, excessive or poorly configured alarm thresholds may generate unnecessary communication traffic.

Utilities should review alarm policies regularly to ensure that notifications remain operationally meaningful.

An alarm that produces frequent false positives consumes both battery energy and operational resources.

Remote Diagnostics Should Be Used Judiciously

Remote diagnostics reduce field visits and improve troubleshooting efficiency.

Typical diagnostic activities include:

- Signal quality checks
- Battery status queries
- Firmware version verification
- Configuration review
- Communication statistics

These functions are valuable but should be used selectively.

Routine polling of large meter populations without a defined operational purpose may increase communication activity without delivering proportional operational benefit.

Firmware Updates Require Battery Planning

Over-the-air firmware updates have become an important feature of modern AMI systems.

They allow manufacturers and utilities to:

- Correct software defects
- Improve cybersecurity
- Introduce new features
- Optimise communication behaviour
- Enhance operational performance

However, firmware downloads represent one of the most energy-intensive activities performed by a battery-powered meter.

Utilities should therefore plan update campaigns carefully by considering:

- Number of devices
- Timing
- Network capacity
- Battery condition
- Operational urgency

Well-managed firmware campaigns preserve both network performance and battery life.

Operational Policies Should Be Reviewed Periodically

AMI deployments evolve over time.

Policies established during the initial rollout may no longer remain appropriate several years later.

Periodic review should consider:

- Reporting frequency
- Alarm configuration
- Diagnostic procedures
- Firmware strategy
- Data retention policies
- Network performance
- Battery health trends

Regular review helps ensure that operational practices continue to support both business objectives and long-term battery performance.

Balance Operational Value Against Battery Consumption

Every additional function consumes some portion of the Battery Life Budget.

The objective is therefore not to minimise communication but to maximise operational value for the energy consumed.

Utilities should ask practical questions such as:

- Does this additional transmission improve operational decision-making?
- Will this alarm reduce water loss or improve customer service?
- Is this diagnostic information genuinely required?
- Can multiple activities be combined into a single communication session?

Thinking in terms of **value per unit of energy** encourages more efficient AMI operation.

Engineering Insight

Battery life is not consumed by the battery alone. It is consumed by every operational decision that requires the meter to wake, process data, and communicate.

Practitioner Observation

During long-term AMI deployments, operational requirements often expand beyond those defined during the original project. Utilities may request more frequent reporting, introduce new alarm functions, or increase remote diagnostics without reassessing their impact on battery consumption. Periodic review of operational policies helps ensure that additional functionality delivers measurable value while remaining consistent with the assumptions of the original Battery Life Budget.

Key Takeaways

- Utility operational policies continue to influence battery life after deployment.
- Reporting frequency is one of the most significant operational variables.
- Alarm strategies should prioritise meaningful events rather than maximum sensitivity.
- Firmware update campaigns should be planned with battery and network capacity in mind.
- Operational value should always be balanced against energy consumption.

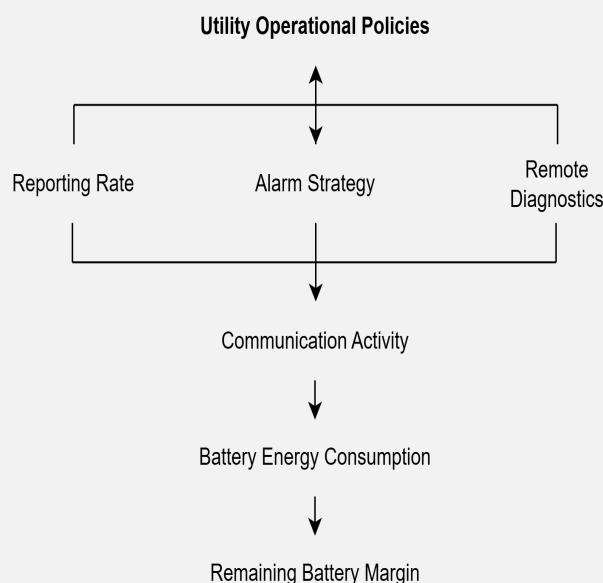


Figure 6.4 — Utility Decisions That Consume the Battery Life Budget

Operational policies influence communication activity throughout the service life of a smart water meter. Managing reporting frequency, alarms, diagnostics, and firmware updates helps preserve the Battery Life Budget while maintaining operational effectiveness.

Practical Utility Checklist — Managing Battery Life During Operations

Utilities can improve long-term battery performance by routinely reviewing the following questions:

Operational Question	Recommended Practice
Is the reporting interval still appropriate?	Align reporting frequency with operational needs rather than technology capability.
Are alarm thresholds generating excessive notifications?	Review alarm performance and eliminate unnecessary events.
Are remote diagnostics being used efficiently?	Perform targeted diagnostics instead of routine mass polling.
Are firmware updates planned and prioritised?	Combine updates where possible and avoid unnecessary campaigns.

Are communication KPIs being monitored?	Track retry rates, data availability, upload latency, and attachment success.
Has the Battery Life Budget been reassessed after operational changes?	Recalculate expected battery impact whenever major policy changes are introduced.

6.5 Monitoring Battery Health During Deployment

Battery qualification provides confidence before deployment.

However, maintaining confidence throughout a ten- to fifteen-year operational life requires continuous monitoring after installation.

Modern AMI systems generate a wealth of operational data that can be used to assess battery health, identify abnormal energy consumption, and support proactive maintenance planning.

Rather than treating battery replacement as a reactive activity, utilities can use operational analytics to manage battery performance throughout the asset lifecycle.

Battery Monitoring Is More Than Battery Voltage

Battery voltage is one indicator of battery condition, but it should not be interpreted in isolation.

Many primary lithium batteries maintain a relatively stable voltage throughout much of their service life before declining rapidly near end-of-life.

Consequently, battery voltage alone may provide limited warning of future battery depletion.

Utilities should therefore monitor battery health using multiple operational indicators rather than relying solely on voltage measurements.

Monitor Operational KPIs

AMI platforms continuously generate operational information that reflects battery performance.

Useful indicators include:

- Daily data availability
- Communication success rate
- Retry frequency
- Upload latency
- Network attachment time

- Missed scheduled reports
- Alarm transmission success
- Battery voltage trend

Monitoring these KPIs together provides a more complete picture of device health than any individual parameter.

Trend Analysis Is More Valuable Than Individual Measurements

A single battery measurement provides only a snapshot in time.

Long-term trends often reveal more meaningful information.

Examples include:

- Gradual increase in communication retries
- Slowly declining battery voltage
- Increasing upload duration
- Rising network attachment time
- Higher frequency of missed reports

Monitoring changes over months or years allows utilities to detect gradual degradation before it becomes a service issue.

Identify Abnormal Devices Early

Not every meter consumes battery energy at the same rate.

Utilities should identify devices that behave differently from the wider population.

Examples include:

- Excessive retry rates
- Unusually frequent communication
- Unexpected battery voltage decline
- Repeated communication failures
- Higher alarm frequency

Early investigation helps determine whether the cause is:

- Battery ageing
- RF coverage deterioration
- Installation issues
- Firmware behaviour
- Environmental conditions

Addressing these issues early can prevent unnecessary battery replacements.

Monitor the Fleet, Not Just Individual Meters

Effective battery management should focus on population-level trends as well as individual devices.

Useful fleet-level indicators include:

- Average battery voltage by region
- Retry distribution
- Daily data availability
- Communication success by operator
- Alarm frequency
- Battery replacement rate

Analysing fleet behaviour enables utilities to identify systematic issues affecting entire deployment areas rather than isolated devices.

Predictive Maintenance Improves Operational Efficiency

Battery monitoring supports predictive rather than reactive maintenance.

Instead of replacing batteries only after communication is lost, utilities can prioritise meters showing early signs of abnormal behaviour.

Benefits include:

- Better maintenance planning
- Reduced emergency site visits
- Lower operational costs
- Improved customer service
- Higher AMI data availability

Predictive maintenance also allows battery replacement to be coordinated with other field activities, improving workforce efficiency.

Analytics Improve Future Battery Life Budgets

Field data should not only support current operations but also improve future engineering designs.

Long-term operational data can refine assumptions relating to:

- Reporting frequency
- Communication retries
- Environmental exposure
- Seasonal behaviour
- Firmware optimisation
- Network performance

This feedback strengthens future Battery Life Budget calculations and contributes to continuous product improvement.

Engineering Insight

The best time to investigate battery problems is before customers notice them. Continuous monitoring transforms battery management from reactive replacement to predictive asset management.

Practitioner Observation

In mature AMI deployments, battery failures rarely occur simultaneously across the entire fleet. Instead, a small number of devices typically begin showing increased retries, reduced data availability, or abnormal battery trends long before widespread issues emerge. Utilities that monitor these indicators can often identify the underlying cause—such as deteriorating RF conditions, firmware behaviour, or installation problems—before replacing batteries unnecessarily.

Key Takeaways

- Battery voltage alone is not a sufficient indicator of battery health.
- Operational KPIs provide valuable insight into long-term battery performance.
- Trend analysis is more informative than isolated measurements.
- Fleet-level monitoring identifies systematic issues across deployments.
- Predictive maintenance improves operational efficiency and customer service.

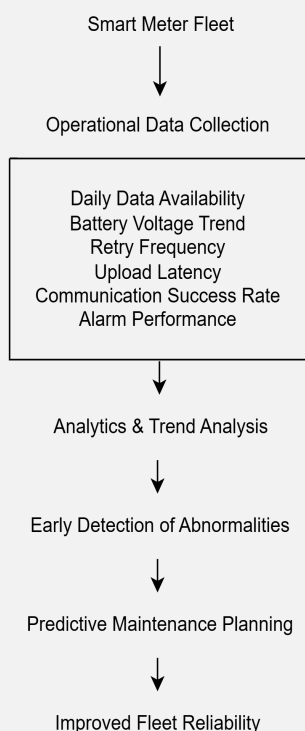


Figure 6.5 — Battery Health Monitoring Throughout the AMI Lifecycle

Continuous monitoring of operational KPIs enables utilities to identify abnormal battery behaviour early, supporting predictive maintenance and improving long-term AMI reliability.

Practical Utility Dashboard

A mature AMI battery monitoring dashboard should include the following KPIs:

KPI	Why It Matters
Daily Data Availability	Detects communication degradation before complete failure.
Monthly Reading Success Rate	Measures overall fleet communication reliability.
Battery Voltage Trend	Indicates long-term battery ageing rather than isolated voltage values.
Average Retry Count	Highlights communication inefficiencies that increase battery consumption.
Upload Latency	Reveals deteriorating network performance or attachment delays.

Low Battery Alarm Rate	Tracks the percentage of meters approaching end-of-life.
Firmware Version Distribution	Identifies whether battery-related issues are linked to specific firmware releases.
Battery Replacement Rate	Supports lifecycle planning and future procurement forecasting.

6.6 Managing Battery Life Throughout the AMI Lifecycle

Battery life should not be viewed as a fixed product specification established during design and forgotten after deployment.

Instead, it should be managed throughout the operational life of the AMI system.

Every stage of the asset lifecycle—from procurement and installation to operation, maintenance, firmware management, and eventual replacement—can influence long-term battery performance.

Utilities that actively manage battery life as part of their asset management strategy are better positioned to maximise return on investment, reduce operational costs, and maintain high levels of data availability throughout the deployment.

Battery Management Begins Before Procurement

Battery life management starts long before the first meter is installed.

During procurement, utilities should define clear operational requirements rather than relying solely on headline battery life claims.

Examples include:

- Required reporting frequency
- Expected daily data availability
- Communication technology
- Deployment environment
- Environmental qualification requirements
- Battery replacement strategy
- Target operational lifetime

Clearly defining these requirements allows manufacturers to develop Battery Life Budgets that align with the intended deployment.

Deployment Should Preserve Design Assumptions

The installation phase should maintain the conditions assumed during engineering qualification.

Important activities include:

- RF coverage verification
- Proper installation practices
- Environmental assessment
- Commissioning tests
- Initial communication validation
- Baseline battery recording

These activities establish the operational foundation for long-term battery performance.

Operation Requires Continuous Review

AMI deployments evolve over many years.

Operational policies should therefore be reviewed periodically to ensure they remain aligned with both business objectives and battery life expectations.

Typical review topics include:

- Reporting frequency
- Alarm strategy
- Firmware management
- Remote diagnostics
- Data collection policies
- Communication performance

Changes introduced during operation should be evaluated for their impact on the Battery Life Budget.

Maintenance Should Be Predictive

Traditional maintenance often responds only after failures occur.

Battery-powered AMI systems benefit from predictive maintenance strategies that use operational data to identify emerging issues before service is affected.

Useful predictive indicators include:

- Increasing retry frequency
- Declining communication success
- Battery voltage trends
- Data availability deterioration
- Abnormal device behaviour

Predictive maintenance enables utilities to optimize field resources while maintaining customer service levels.

Firmware Is Part of Lifecycle Management

Firmware evolves throughout the operational life of the meter.

Updates may introduce:

- New functionality
- Cybersecurity improvements
- Communication optimisation
- Bug fixes
- Regulatory compliance updates

Each firmware release should be evaluated for its impact on battery consumption.

Firmware lifecycle management should therefore include:

- Power consumption assessment
- Controlled deployment
- Post-update monitoring
- Rollback capability where appropriate

Battery life should remain a design consideration throughout software evolution.

Use Field Data to Improve Future Projects

Every deployed AMI system generates valuable operational knowledge.

Utilities should use this information to improve:

- Future procurement specifications
- Battery Life Budget assumptions
- Communication policies
- Installation standards
- Firmware design
- Maintenance planning

Lessons learned from one deployment strengthen the success of future projects.

Plan for End-of-Life

Even with excellent engineering and operational management, batteries have a finite service life.

Utilities should therefore prepare well in advance for end-of-life activities.

Planning should consider:

- Expected replacement timeline
- Geographic replacement priorities
- Workforce availability
- Spare inventory
- Customer communication
- Budget allocation
- Meter replacement versus battery replacement

Proactive planning reduces operational disruption and supports smoother asset renewal programmes.

Battery Life Is an Asset Management KPI

Battery performance should be incorporated into broader asset management frameworks alongside:

- Data availability
- Meter accuracy
- Communication reliability
- Asset utilisation
- Maintenance efficiency
- Lifecycle cost

Treating battery life as an operational KPI encourages cross-functional collaboration between engineering, operations, IT, and asset management teams.

Engineering Insight

Battery life is not simply an engineering outcome—it is a lifecycle management responsibility shared by manufacturers, utilities, and system operators throughout the operational life of the AMI system.

Practitioner Observation

Utilities with mature AMI programmes rarely manage battery life as an isolated maintenance issue. Instead, they integrate battery performance into broader asset management practices, using operational analytics, communication KPIs, firmware governance, and lifecycle planning to optimise the performance of the entire meter population. This approach supports both operational reliability and long-term investment value.

Key Takeaways

- Battery management begins during procurement and continues throughout the AMI lifecycle.
- Deployment practices should preserve the assumptions used during engineering qualification.
- Operational policy changes should be evaluated for their impact on battery consumption.
- Predictive maintenance improves efficiency and reduces unexpected failures.
- Field experience should continuously improve future AMI deployments.



Figure 6.6 — Battery Lifecycle Management Framework

Battery life should be managed as a continuous lifecycle process. Lessons learned during operation and maintenance should inform future procurement, engineering, and deployment strategies, creating a cycle of continuous improvement.

Practical Utility Maturity Model

A useful addition to this part of the whitepaper is a maturity model that helps utilities assess their battery lifecycle management capability.

Maturity Level	Characteristics
Level 1 – Reactive	Batteries are replaced only after communication is lost.
Level 2 – Monitored	Battery voltage and communication KPIs are tracked periodically.
Level 3 – Managed	Battery trends influence maintenance planning and operational policies.
Level 4 – Predictive	Analytics identify abnormal battery behaviour before failures occur.
Level 5 – Optimised	Battery lifecycle management is fully integrated with asset management, procurement, firmware governance, and continuous improvement.

This maturity model provides readers with a practical benchmark and reinforces the whitepaper's core message that **battery life is a managed asset, not merely a component specification**.

6.7 Common Field Mistakes That Shorten Battery Life

Battery life rarely decreases because of a single catastrophic event.

In most deployments, reduced operational lifetime results from the gradual accumulation of small decisions that collectively increase battery consumption over many years.

These decisions may involve installation practices, communication policies, firmware management, maintenance procedures, or operational governance.

Recognising these common field mistakes allows utilities to preserve the Battery Life Budget established during product development and maximise the operational value of their AMI investment.

Mistake 1 — Increasing Reporting Frequency Without Reassessing Battery Impact

Operational requirements often evolve after deployment.

Utilities may decide to:

- Increase reporting frequency

- Collect additional consumption data
- Enable more diagnostic information
- Expand alarm functionality

While these changes provide greater operational visibility, they also increase communication activity.

Before modifying reporting policies, utilities should evaluate how the additional communication affects the Battery Life Budget.

Mistake 2 — Ignoring Communication KPIs

Many utilities monitor meter readings but pay less attention to communication performance.

Important indicators such as:

- Retry frequency
- Upload latency
- Communication success rate
- Daily data availability

provide early warning of emerging battery issues.

Ignoring these indicators allows communication inefficiencies to persist, gradually increasing battery consumption across the deployment.

Mistake 3 — Treating All Installations the Same

Deployment environments differ significantly.

For example:

- Underground chambers
- High-rise buildings
- Rural installations
- Industrial facilities
- Flood-prone areas

Each environment presents different communication and environmental challenges.

Applying identical installation procedures across all deployment scenarios may reduce long-term battery performance.

Installation standards should be adapted to local operating conditions.

Mistake 4 — Delaying Firmware Maintenance

Firmware improvements often include:

- Communication optimisation
- Power management enhancements
- Security updates
- Bug fixes

Delaying important firmware updates may prevent deployed meters from benefiting from improved battery efficiency.

Conversely, excessive or poorly planned firmware campaigns may also consume unnecessary battery energy.

Firmware management therefore requires careful planning throughout the operational lifecycle.

Mistake 5 — Investigating Batteries Before Investigating the System

When battery life appears shorter than expected, the battery is often assumed to be the cause.

However, engineers should first evaluate:

- Communication quality
- Retry behaviour
- RF coverage
- Firmware configuration
- Installation conditions
- Operational policies

Many apparent battery issues originate elsewhere within the AMI system.

Understanding the root cause helps avoid unnecessary battery replacements.

Mistake 6 — Treating Battery Replacement as the Only Maintenance Strategy

Replacing batteries addresses the symptom but may not resolve the underlying cause of increased energy consumption.

For example:

- Poor RF coverage
- Communication retries
- Water ingress
- Incorrect firmware settings
- Installation deficiencies

If these conditions remain unchanged, replacement batteries may experience the same shortened operational life.

Maintenance activities should therefore include investigation as well as replacement.

Mistake 7 — Failing to Learn From Operational Data

Every AMI deployment generates valuable engineering knowledge.

Utilities that fail to analyse operational data miss opportunities to improve:

- Future procurement
- Installation practices
- Firmware development
- Communication policies
- Maintenance planning
- Battery Life Budget assumptions

Continuous learning transforms operational experience into improved long-term system performance.

Build a Culture of Continuous Improvement

Long-life battery performance depends not only on technology but also on organisational learning.

Utilities should establish processes to:

- Review communication KPIs
- Analyse battery trends
- Investigate abnormal devices
- Update installation procedures
- Refine operational policies
- Share lessons across departments

This continuous improvement approach helps ensure that each deployment performs better than the previous one.

Engineering Insight

Most battery life reductions are not caused by one major mistake. They result from many small operational decisions that gradually consume the Battery Life Budget over time.

Practitioner Observation

Experienced utilities rarely view unexpected battery consumption as an isolated maintenance issue. Instead, they investigate the wider operational context, including communication performance, installation quality, firmware behaviour, and changing business requirements. This systems-based approach frequently identifies opportunities

to improve both battery life and overall AMI performance without increasing hardware costs.

Key Takeaways

- Operational policy changes should always be evaluated against their battery impact.
- Communication KPIs are leading indicators of future battery performance.
- Installation practices should reflect local deployment conditions.
- Firmware management should balance operational benefits with energy consumption.
- Root-cause analysis is more valuable than simply replacing batteries.
- Continuous learning improves future AMI deployments and Battery Life Budget accuracy.

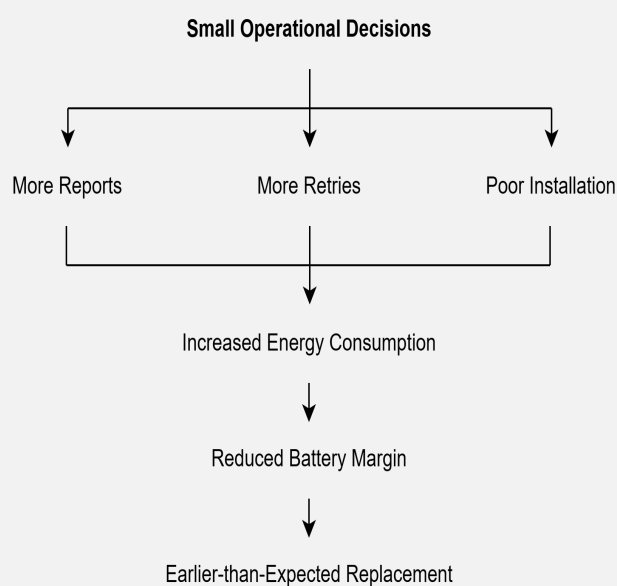


Figure 6.7 — Small Decisions, Large Lifecycle Impact

Long-term battery life is influenced by the cumulative effect of many operational decisions. Individually these changes may appear minor, but together they can significantly reduce the remaining Battery Life Budget over the service life of the meter.

Practical Utility Self-Assessment

Utilities can periodically review the following questions to evaluate whether operational practices are supporting long-term battery performance:

Assessment Question	Why It Matters
Have reporting intervals changed since deployment?	Higher reporting frequency increases communication energy consumption.
Are retry rates being monitored and investigated?	Persistent retries often indicate RF or network issues before battery problems become visible.
Are installation standards reviewed using field experience?	Lessons learned improve future deployment quality.
Are firmware updates assessed for their power impact?	New features should not unintentionally increase energy consumption.
Is battery replacement supported by root-cause analysis?	Replacing batteries without resolving underlying issues may repeat the same failures.
Are operational lessons incorporated into future procurements?	Continuous improvement strengthens future AMI deployments and procurement specifications.

Summary

A Battery Life Budget provides an engineering prediction of expected service life under defined operating assumptions.

However, once a smart water meter is deployed, actual battery performance becomes influenced by a much broader range of operational factors that extend beyond product design.

Throughout this chapter, we have explored how communication quality, installation practices, utility operational policies, maintenance strategies, and continuous monitoring collectively determine whether deployed meters achieve their intended operational lifetime.

One of the most important lessons is that battery life should not be regarded as a fixed specification.

Instead, it should be managed as a dynamic operational resource throughout the entire lifecycle of the AMI system.

Network conditions evolve.

Firmware changes.

Utility requirements expand.

Environmental conditions vary.

Operational policies change.

Each of these factors influences how quickly the available Battery Life Budget is consumed.

Utilities that actively monitor communication performance, review operational policies, maintain installation quality, and analyse fleet-wide battery trends are significantly better positioned to maximise asset lifetime than organisations relying solely on initial product specifications.

Equally important is the recognition that battery management is a shared responsibility.

Manufacturers establish the engineering foundation through sound product design and qualification.

System integrators ensure correct deployment.

Network operators provide reliable communication infrastructure.

Utilities manage daily operational policies and long-term asset performance.

Success depends on the effective coordination of all these stakeholders.

Ultimately, long battery life is not achieved through battery chemistry alone.

It results from disciplined engineering, disciplined operations, and continuous lifecycle management working together over many years.

Key Message

The principal lessons from this chapter include:

- Actual battery life depends on both engineering assumptions and real-world operating conditions.
- Communication quality is one of the most significant operational influences on battery consumption.
- Installation quality affects RF performance, environmental exposure, and long-term reliability.
- Utility operational policies directly influence communication activity and battery usage.
- Battery health should be monitored using operational KPIs rather than battery voltage alone.
- Predictive maintenance improves operational efficiency and reduces unexpected failures.
- Continuous improvement enables future deployments to benefit from operational experience.

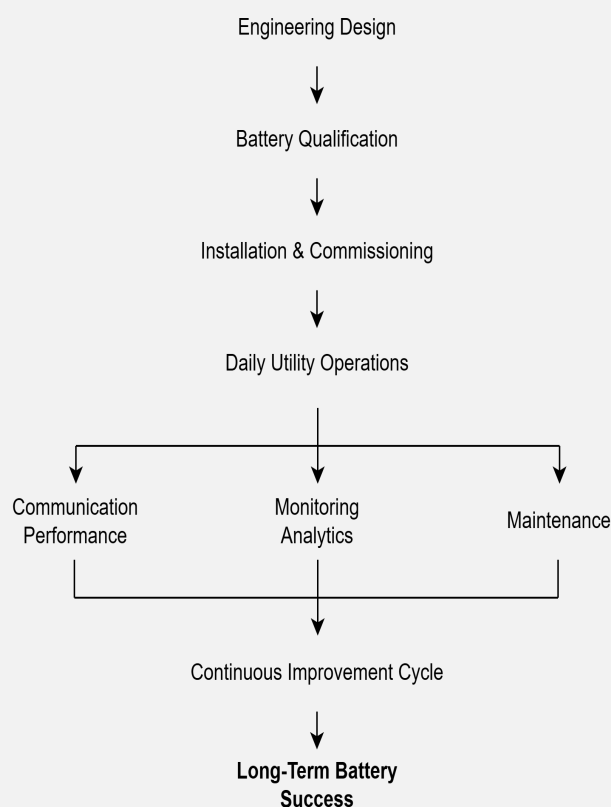


Figure 6.8 — Battery Lifecycle Management in Practice

Long-term battery performance is achieved through continuous lifecycle management. Engineering design, quality installation, operational monitoring, maintenance, and continuous improvement work together to maximise the value of the original Battery Life Budget.

Practical Framework — The Battery Lifecycle Management Cycle

The discussions in Chapters 5 and 6 can be summarised into a practical management cycle for utilities:

Lifecycle Stage	Primary Objective	Key Activities
Design	Build an accurate Battery Life Budget	Power optimisation, battery modelling, qualification testing
Procurement	Define measurable battery requirements	KPIs, reporting assumptions, environmental requirements
Deployment	Preserve engineering assumptions	RF validation, installation quality, commissioning tests

Operation	Manage battery consumption	Reporting policies, alarm strategy, firmware governance
Monitoring	Detect abnormal behaviour early	Battery analytics, communication KPIs, trend analysis
Maintenance	Optimise lifecycle cost	Predictive maintenance, root-cause analysis, targeted replacement
Continuous Improvement	Improve future deployments	Feedback into engineering, procurement, and operational policies

This framework demonstrates that **battery management is not a single event, but a continuous process spanning the entire AMI lifecycle.**

Disclaimer

The views and observations expressed in this paper are the author's own and are intended for general industry discussion and educational purposes. They are based on publicly available research, practical engineering experience, and regional industry observations, and do not represent the official position, product commitments, customer relationships, or commercial practices of any employer, client, utility, or affiliated organisation. Utilities should evaluate deployment decisions within their own operational, technical, regulatory, and procurement context.

About the Author



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Vincent Wong brings over **25 years of engineering and product leadership experience** across industrial infrastructure and technology sectors, with the past decade focused on smart metering, Advanced Metering Infrastructure (AMI), and NB-IoT deployment strategies in ASEAN utility environments. As a senior engineering and product development executive, his work spans hardware engineering, connectivity systems, and utility digitalization initiatives. His writing focuses on the practical operational, connectivity, deployment, and governance realities of smart utility programs in Southeast Asia.

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